Localization and Synchronization of a Mobile User Aided by Reconfigurable Intelligent Surfaces with Uncertain Positions

B.Sc. Project Report — English Summary

Mohammad Amin Khoshkam

Department of Electrical Engineering, Sharif University of Technology, Tehran, Iran

Supervisor: Dr. Rouhollah Amiri Project course instructor: Dr. Bijan Vosoughi Vahdat

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This document is a condensed English version of the original B.Sc. report, which was written in Persian. It keeps the problem formulation, the main equations, the numerical results and the conclusions; derivation details and the glossary are omitted. All numbers are taken from the CSV outputs in `crlb/crlb_results/` and `estimator/hybrid_map_results/`.

Abstract

We jointly estimate the 3-D position and velocity of a mobile user equipment (UE), the relative frequency error and the clock offset, and the positions, velocities and complex path gains associated with reconfigurable intelligent surfaces (RISs). The base-station (BS) position is known, while the RIS geometry is available only through a Gaussian prior; instead of assuming perfectly known RISs, the geometric uncertainty is built directly into the localization, synchronization and estimation problem. The BS transmits OFDM pilots over several subcarriers and time snapshots, and the received-signal model gathers time-varying delay, Doppler, frequency error and clock offset into a single *effective delay*. To obtain the fundamental accuracy limit, the data information is combined with the RIS prior to form a hybrid Fisher information matrix (FIM); the complex path gains are treated as nuisance parameters and removed with a Schur complement. We then propose a hybrid maximum-a-posteriori / maximum-likelihood (MAP/ML) estimator in which only the RIS states use prior information, while the UE, synchronization and channel parameters do not. Because the model is linear in the path gains, these are eliminated analytically by variable projection; coarse-to-fine synchronization search, random multi-start and local constrained optimization reduce sensitivity to initialization. With three RISs at 20 dB SNR, the position and velocity error bounds of the UE are 1.176 m and 1.864 m/s, and the bounds on the frequency error and clock offset are 0.00146 ppm and 0.985 ns. In a numerical run, the estimator reduces the UE position error from 3.212 m to 0.617 m with a normalized data-fit cost of 0.999 per sample. The results show that localization and synchronization accuracy depends strongly on the quality of the RIS prior, and that even with a full-rank FIM a poor geometry can produce a large condition number and noticeably degrade accuracy.

Keywords: reconfigurable intelligent surface; hybrid Cramér–Rao bound; MAP estimation; variable projection; joint localization and synchronization; geometric uncertainty.

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1 Introduction

1.1 Problem statement

RISs create controllable reflected paths that extend coverage and provide a better geometry for localization, which is especially valuable when the line-of-sight (LoS) path between the BS and the UE is weak or blocked. However, an RIS can act as a reliable *geometric anchor* only if its position and motion are known accurately. In practice, surveying and installation errors, structural displacement, vibration and calibration errors mean that the effective RIS position — and even its velocity — differs from the nominal values. Treating these quantities as perfectly known leads to over-optimistic models and biased estimates; treating them as completely unknown increases the number of unknowns and may make the problem unidentifiable.

This project considers the middle ground. The BS position $\boldsymbol{q} \in \mathbb{R}^3$ is known. A mobile UE has an initial position $\boldsymbol{u}_0$ and constant velocity $\dot{\boldsymbol{u}}$, both unknown. K RISs have initial positions $\boldsymbol{p}_{k,0}$ and velocities $\dot{\boldsymbol{p}}_k$ that are not known exactly; their uncertainty is described by a Gaussian prior. In addition, the relative frequency error η, the clock offset δ_c and the complex path gains β_k are unknown. The BS transmits known OFDM pilots on N subcarriers over L time snapshots: multiple subcarriers carry delay information, while multiple snapshots reveal the changes caused by UE motion, RIS motion and Doppler.

Two questions are addressed:

1. Given unknown channel gains and uncertain RIS geometry, what is the fundamental accuracy achievable for UE position, velocity and synchronization? This is answered with the hybrid FIM and hybrid Cramér–Rao bound.

2. How can an estimator exploit the available RIS prior without imposing artificial priors on the UE, synchronization and channel parameters? This leads to the hybrid MAP/ML estimator.

1.2 Technical challenges

- The phase of each observation is a periodic, nonlinear function of the effective delay, so the cost function has many local minima and a local optimizer may not converge from a poor initial point.

- The frequency error and clock offset are coupled with propagation delay and motion: the clock offset acts like an extra delay, while frequency error and relative motion both create time-dependent phase changes. A single snapshot cannot separate them; multiple snapshots are required.

- The model is linear in the path gains β_k but nonlinear in the geometric and synchronization parameters. Searching jointly over all of them increases the dimension of the nonlinear problem.

- Adding RIS positions and velocities as unknowns without any prior can make the FIM rank-deficient; an over-confident prior, on the other hand, underestimates the true installation uncertainty.

- A full-rank FIM does not guarantee numerical stability: a very large condition number means that some directions in parameter space carry very little information.

1.3 Contributions

1. A discrete OFDM signal model for the direct path and K RIS paths, including UE motion, RIS motion, propagation delay, Doppler, frequency error and clock offset.

2. A unified parameter vector (UE state, synchronization, RIS states, complex gains), with an explicit mapping between its conceptual ordering and the storage order in the code.

3. The data FIM, with prior information added *only* to the RIS position and velocity blocks, so that no artificial prior is placed on UE, synchronization or channel parameters.

4. Elimination of the channel gains as nuisance parameters with a Schur complement, and derivation of the position error bound (PEB), velocity error bound (VEB) and synchronization bounds.

5. A hybrid MAP/ML estimator that jointly uses the RIS prior and the observations.

6. Variable projection to remove the channel gains analytically from the nonlinear search.

7. Coarse-to-fine search over the synchronization parameters, random multi-start and local constrained optimization to reduce sensitivity to initialization.

8. Numerical study of the bounds and the estimator, of the effect of RIS prior quality, and of the effect of geometry on rank, condition number and achievable accuracy.

1.4 Assumptions and scope

- UE and RIS motion follow a constant-velocity model over the observation window (acceleration neglected).
- Paths are resolvable, and the complex path gains are constant over the observation block; pilots are perfectly known, and the post-processing noise is circularly symmetric complex white Gaussian.
- Inter-carrier interference (ICI) is neglected: the normalized frequency error and residual Doppler are assumed small compared with the subcarrier spacing, so their main effect is captured by phase and effective delay.
- Prior information exists only for the RIS states. Because part of the parameter vector is random (with a prior) and part is deterministic, the framework is *hybrid* — neither fully Bayesian nor fully classical.

2 System and Signal Model

2.1 Geometry and motion

Subcarrier indices are symmetric around the centre subcarrier, $\mathcal{N} = \{-\frac{N-1}{2}, \dots, 0, \dots, \frac{N-1}{2}\}$ for odd N, and the snapshot times are centred around zero,

$$T_l = \Big(l - \frac{L+1}{2}\Big)T_s, \qquad l = 1, \dots, L, \tag{1}$$

where T_s is the snapshot interval. Centring the time axis is not cosmetic: it reduces the numerical correlation between initial position and velocity, so the two are better separated. With constant velocities,

$$\boldsymbol{u}_l = \boldsymbol{u}_0 + \dot{\boldsymbol{u}}\, T_l, \qquad \boldsymbol{p}_{k,l} = \boldsymbol{p}_{k,0} + \dot{\boldsymbol{p}}_k\, T_l, \quad k = 1, \dots, K. \tag{2}$$

Path $k = 0$ is the direct BS–UE path; paths $k = 1, \dots, K$ go from the BS to RIS k and then to the UE. Each RIS creates one effective reflected path; its internal array response and phase profile are absorbed into β_k, so the focus stays on position, motion and synchronization.

2.2 Delay and path rate

The delays of the direct and reflected paths are

$$\tau_{l,0} = \frac{1}{c}\|\boldsymbol{u}_l - \boldsymbol{q}\|, \qquad \tau_{l,k} = \frac{1}{c}(\|\boldsymbol{q} - \boldsymbol{p}_{k,l}\| + \|\boldsymbol{p}_{k,l} - \boldsymbol{u}_l\|), \tag{3}$$

and, with the sign convention used in the implementation, the geometric path rates are

$$\nu_{l,0} = \frac{1}{c}\,\frac{\dot{\boldsymbol{u}}^{\mathsf{T}}(\boldsymbol{u}_l - \boldsymbol{q})}{\|\boldsymbol{u}_l - \boldsymbol{q}\|}, \tag{4}$$

$$\nu_{l,k} = \frac{1}{c}\left(-\frac{\dot{\boldsymbol{p}}_k^{\mathsf{T}}(\boldsymbol{q} - \boldsymbol{p}_{k,l})}{\|\boldsymbol{q} - \boldsymbol{p}_{k,l}\|} + \frac{(\dot{\boldsymbol{p}}_k - \dot{\boldsymbol{u}})^{\mathsf{T}}(\boldsymbol{p}_{k,l} - \boldsymbol{u}_l)}{\|\boldsymbol{p}_{k,l} - \boldsymbol{u}_l\|}\right). \tag{5}$$

Sign conventions for path rate and Doppler differ between references; what matters is that the definition of $\nu_{l,k}$, the effective delay and the signal phase are consistent throughout the derivations and the code.

2.3 Clock offset, frequency error and effective delay

The UE and BS oscillators are related by $f_{\mathrm{UE}} = (1+\eta) f_{\mathrm{BS}}$, where η is the (dimensionless) relative frequency error, and δ_c is the clock offset between the two time references. Linearizing the path variations over the observation window, all effects are gathered into one effective delay:

$$a_{l,k} = \tau_{l,k} - (\nu_{l,k} - \eta)\, T_l - \delta_c. \tag{6}$$

The estimator never observes these effects separately but only through $a_{l,k}$; separating them requires sufficient frequency *and* time diversity. In the default `reference-linearized` geometry model, $\tau_{l,k}$ and $\nu_{l,k}$ are evaluated at the reference time and motion enters through the T_l term.

2.4 Received signal

After cyclic-prefix removal and the DFT, the sample on subcarrier n in snapshot l is

$$y_{n,l} = \sqrt{\frac{\varepsilon}{N}}\, x_{n,l} \sum_{k=0}^{K} \beta_k \exp(-j2\pi n \Delta f\, a_{l,k}) + z_{n,l}, \qquad z_{n,l} \sim \mathcal{CN}(0, \sigma^2), \tag{7}$$

where $x_{n,l}$ is the known pilot, ε the pilot energy and Δf the subcarrier spacing. Stacking all samples as $\boldsymbol{y} = \mathrm{vec}\{y_{n,l}\} \in \mathbb{C}^{NL}$ and defining the design matrix $\boldsymbol{A}(\boldsymbol{\theta}) \in \mathbb{C}^{NL \times (K+1)}$ with entries

$$[\boldsymbol{A}(\boldsymbol{\theta})]_{(n,l),k} = \sqrt{\varepsilon/N}\, x_{n,l} \exp(-j2\pi n \Delta f\, a_{l,k}(\boldsymbol{\theta})), \tag{8}$$

the model becomes

$$\boldsymbol{y} = \boldsymbol{A}(\boldsymbol{\theta})\, \boldsymbol{\beta} + \boldsymbol{z}, \qquad \boldsymbol{\beta} = [\beta_0, \ldots, \beta_K]^{\mathsf{T}}. \tag{9}$$

Model (9) is the key relation of the project: $\boldsymbol{A}$ depends nonlinearly on position, velocity and synchronization, but the model is *linear* in $\boldsymbol{\beta}$. For a fixed $\boldsymbol{\theta}$, the optimal $\boldsymbol{\beta}$ can therefore be computed in closed form.

2.5 Parameter vector

The physical parameters are stored in the MATLAB code as

$$\boldsymbol{\theta} = [\boldsymbol{u}_0^{\mathsf{T}},\ \dot{\boldsymbol{u}}^{\mathsf{T}},\ \boldsymbol{p}_{1,0}^{\mathsf{T}}, \ldots, \boldsymbol{p}_{K,0}^{\mathsf{T}},\ \dot{\boldsymbol{p}}_1^{\mathsf{T}}, \ldots, \dot{\boldsymbol{p}}_K^{\mathsf{T}},\ \eta,\ \delta_c]^{\mathsf{T}} \in \mathbb{R}^{8+6K}. \tag{10}$$

For the statistical analysis they are grouped by role: the UE and synchronization parameters $\boldsymbol{\psi} = [\boldsymbol{u}_0^{\mathsf{T}}, \dot{\boldsymbol{u}}^{\mathsf{T}}, \eta, \delta_c]^{\mathsf{T}} \in \mathbb{R}^8$ (no prior) and the RIS states $\boldsymbol{\rho} \in \mathbb{R}^{6K}$ (with prior), extracted with selection matrices $\boldsymbol{\psi} = \boldsymbol{E}_\psi \boldsymbol{\theta}$, $\boldsymbol{\rho} = \boldsymbol{E}_\rho \boldsymbol{\theta}$. For a real-valued FIM, the complex gains are split into real and imaginary parts,

$$\boldsymbol{\gamma} = [\boldsymbol{\theta}^{\mathsf{T}},\ \mathrm{Re}\{\boldsymbol{\beta}\}^{\mathsf{T}},\ \mathrm{Im}\{\boldsymbol{\beta}\}^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^{10+8K}. \tag{11}$$

For $K = 3$: $\dim(\boldsymbol{\theta}) = 26$ and $\dim(\boldsymbol{\gamma}) = 34$.

3 Hybrid Fisher Information and Estimation Bounds

3.1 Data information

The noiseless mean is $\boldsymbol{\mu}(\boldsymbol{\gamma}) = \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{\beta}$. For complex white Gaussian noise with covariance $\sigma^2\boldsymbol{I}$, the FIM entries are [1]

$$[\boldsymbol{J}_{\text{data}}]_{i,j} = \frac{2}{\sigma^2}\sum_{l=1}^{L}\sum_{n\in\mathcal{N}} \text{Re}\left\{\left(\frac{\partial \mu_{n,l}}{\partial \gamma_i}\right)^* \frac{\partial \mu_{n,l}}{\partial \gamma_j}\right\}, \tag{12}$$

with

$$\frac{\partial \mu_{n,l}}{\partial \theta_i} = \sqrt{\frac{\varepsilon}{N}}\, x_{n,l} \sum_{k=0}^{K} \beta_k e^{-j2\pi n\Delta f a_{l,k}} \left(-j2\pi n\Delta f\right) \frac{\partial a_{l,k}}{\partial \theta_i}, \qquad \frac{\partial a_{l,k}}{\partial \eta} = T_l, \quad \frac{\partial a_{l,k}}{\partial \delta_c} = -1. \tag{13}$$

Geometric derivatives use $\nabla_{\boldsymbol{r}}\|\boldsymbol{r}\| = \boldsymbol{w}$ and $\nabla_{\boldsymbol{r}}\boldsymbol{w} = \boldsymbol{P}_{\boldsymbol{w}}^{\perp}/d$ with $\boldsymbol{w} = \boldsymbol{r}/\|\boldsymbol{r}\|$, $\boldsymbol{P}_{\boldsymbol{w}}^{\perp} = \boldsymbol{I} - \boldsymbol{w}\boldsymbol{w}^{\mathsf{T}}$. In the implementation, position and velocity derivatives are computed with central finite differences and checked against simpler analytic relations to avoid sign errors in long expressions.

3.2 RIS prior and the hybrid FIM

The RIS states follow $\boldsymbol{\rho} \sim \mathcal{N}(\bar{\boldsymbol{\rho}}, \boldsymbol{Q}^{-1})$ with the isotropic precision

$$\boldsymbol{Q} = \text{blkdiag}\Big(\tfrac{1}{\sigma_p^2}\boldsymbol{I}_{3K},\ \tfrac{1}{\sigma_v^2}\boldsymbol{I}_{3K}\Big), \qquad \boldsymbol{J}_{\text{prior},\theta} = \boldsymbol{E}_{\rho}^{\mathsf{T}}\boldsymbol{Q}\boldsymbol{E}_{\rho}, \tag{14}$$

and the hybrid FIM is $\boldsymbol{J}_{\gamma} = \boldsymbol{J}_{\text{data}} + \boldsymbol{J}_{\text{prior}}$, where the prior blocks for UE, synchronization and channel parameters are zero.

Three cases must be distinguished: a parameter that is *exactly known* is not part of the unknown vector; a parameter that is *uncertain but has a prior* must stay in the vector with its prior added (dropping it would hide its effect on the UE estimate and yield an over-optimistic bound); and a pure *nuisance* parameter must be removed with a Schur complement, not by deleting its row and column. Here only the channel gains are nuisance parameters.

3.3 Removing the channel gains

Partitioning $\boldsymbol{J}_{\gamma}$ into the physical block and the block of $\boldsymbol{b} = [\text{Re}\{\boldsymbol{\beta}\}^{\mathsf{T}}, \text{Im}\{\boldsymbol{\beta}\}^{\mathsf{T}}]^{\mathsf{T}}$, the equivalent information on the physical parameters is

$$\boldsymbol{J}_e = \boldsymbol{J}_{\theta\theta} - \boldsymbol{J}_{\theta b}\boldsymbol{J}_{bb}^{\dagger}\boldsymbol{J}_{b\theta}. \tag{15}$$

Using $\boldsymbol{J}_{\theta\theta}$ alone would amount to assuming known channel gains and would give an over-optimistic bound.

3.4 Reported bounds

If $\boldsymbol{J}_e$ is full rank, any locally unbiased estimator satisfies $\boldsymbol{C}_{\theta} \succeq \boldsymbol{J}_e^{-1}$, and the reported metrics are

$$\text{PEB}_{\text{UE}} = \sqrt{\text{tr}([\boldsymbol{C}_{\theta}]_{\boldsymbol{u}_0,\boldsymbol{u}_0})}, \quad \text{VEB}_{\text{UE}} = \sqrt{\text{tr}([\boldsymbol{C}_{\theta}]_{\dot{\boldsymbol{u}},\dot{\boldsymbol{u}}})}, \quad \sigma_{\eta,\text{LB}} = \sqrt{[\boldsymbol{C}_{\theta}]_{\eta,\eta}}, \quad \sigma_{\delta,\text{LB}} = \sqrt{[\boldsymbol{C}_{\theta}]_{\delta_c,\delta_c}}, \tag{16}$$

with analogous $\text{PEB}_{\text{RIS},k}$ and $\text{VEB}_{\text{RIS},k}$. The frequency-error bound is reported in ppm ($\times 10^6$) and the clock-offset bound in ns ($\times 10^9$).

3.5 Rank, identifiability and condition number

Local identifiability of all physical parameters requires $\text{rank}(\boldsymbol{J}_e) = 8 + 6K$. A pseudoinverse keeps the computation running in the rank-deficient case but creates no information that is not in the data. Even with full rank, the condition number $\kappa(\boldsymbol{J}_e) = \lambda_{\max}/\lambda_{\min}$ must be checked: a large value means some directions carry far less information than others, making the problem sensitive to noise, scaling and the starting point.

4 Hybrid MAP/ML Estimator

4.1 Joint cost function

$\boldsymbol{\psi}$ and $\boldsymbol{\beta}$ are deterministic unknowns estimated by ML; $\boldsymbol{\rho}$ has a prior and is estimated by MAP. With the likelihood $p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\beta}) \propto \exp(-\|\boldsymbol{y} - \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{\beta}\|_2^2/\sigma^2)$ and the Gaussian prior on $\boldsymbol{\rho}$,

$$(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\beta}}) = \arg \min_{\boldsymbol{\theta} \in \Theta,\, \boldsymbol{\beta}} \left[\frac{\|\boldsymbol{y} - \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{\beta}\|_2^2}{\sigma^2} + \frac{1}{2}(\boldsymbol{\rho} - \bar{\boldsymbol{\rho}})^{\mathsf{T}} \boldsymbol{Q} (\boldsymbol{\rho} - \bar{\boldsymbol{\rho}}) \right], \tag{17}$$

where Θ holds the physical constraints. The prior term must enter with a *positive* sign (it comes from the negative log of a Gaussian); a negative sign would reward distance from the prior mean and make the cost unbounded below. In the limits, $\sigma_p, \sigma_v \to \infty$ gives a pure ML estimator, while $\sigma_p, \sigma_v \to 0$ pins the RISs to their prior mean.

4.2 Variable projection

For fixed $\boldsymbol{\theta}$, the gains solve a linear least-squares problem, $\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}) = \boldsymbol{A}^{\dagger}(\boldsymbol{\theta})\boldsymbol{y}$ (computed with an SVD rather than an explicit inverse). With $\boldsymbol{P}_{\boldsymbol{A}}^{\perp} = \boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^{\dagger}$, the concentrated problem involves only the physical parameters [3]:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta} \in \Theta} J_c(\boldsymbol{\theta}), \qquad J_c(\boldsymbol{\theta}) = \frac{\|\boldsymbol{P}_{\boldsymbol{A}(\boldsymbol{\theta})}^{\perp} \boldsymbol{y}\|_2^2}{\sigma^2} + \frac{1}{2}(\boldsymbol{\rho} - \bar{\boldsymbol{\rho}})^{\mathsf{T}} \boldsymbol{Q} (\boldsymbol{\rho} - \bar{\boldsymbol{\rho}}). \tag{18}$$

For $K = 3$ this removes four complex gains (eight real variables) from the nonlinear search. The $NL \times NL$ projector is never formed: the residual is computed as $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{A}\hat{\boldsymbol{\beta}}$, which matters for $NL = 4128$. Using $\boldsymbol{A}^{\mathsf{H}}\boldsymbol{r} = \boldsymbol{0}$, the gradient is

$$\nabla_{\theta} J_c = -\frac{2}{\sigma^2} \operatorname{Re}\{\boldsymbol{D}_{\theta}^{\mathsf{H}} \boldsymbol{r}\} + \boldsymbol{E}_{\rho}^{\mathsf{T}} \boldsymbol{Q} (\boldsymbol{\rho} - \bar{\boldsymbol{\rho}}), \qquad \boldsymbol{D}_{\theta} = \frac{\partial [\boldsymbol{A}(\boldsymbol{\theta})\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})]}{\partial \boldsymbol{\theta}^{\mathsf{T}}}, \tag{19}$$

with finite-difference steps of 10^{-3} m for positions and 10^{-2} m/s for velocities.

4.3 Scaling and constraints

Parameters differ by many orders of magnitude (metres, m/s, $\eta \sim 10^{-6}$, $\delta_c \sim 10^{-9}$ s), so the optimizer works on scaled variables with $\boldsymbol{S} = \operatorname{diag}(1, \ldots, 1, 10^{-6}, 10^{-9})$. Global bounds are $[0, 25]$ m for the horizontal UE position, $[0, 10]$ m for height, $[-5, 5]$ m/s for velocity components, $[-20, 20]$ ppm for η and $[-300, 300]$ ns for δ_c. Around each starting point a local box is added to keep the solver away from distant periodic solutions; an active local bound in the final solution indicates that the box may be too small.

4.4 Initialization and optimization

Much of the multimodality comes from η and δ_c, so they are initialized in two stages: a coarse grid of 25×33 points over (η, δ_c), then a 13×13 local grid around the best coarse point. At every grid point the gains are re-estimated by least squares, so points are compared at their best possible channel fit. The full algorithm is:

1. build $\boldsymbol{y}$, the centred times and the pilot matrix;
2. set the mean and precision of the RIS prior;
3. run the coarse and fine search over (η, δ_c) on the concentrated cost;
4. generate random starting points (Gaussian perturbations around the initial estimate), apply the constraints and sort them by cost;

5. run a constrained SQP solver (`fmincon`) from the eight best points, with at most 600 iterations and 30 000 function evaluations each;

6. keep the solution with the lowest final cost and recover $\hat{\boldsymbol{\beta}}$;

7. compute the residual, check active local bounds, the FIM rank and the condition number.

A useful diagnostic is the normalized residual $\|\boldsymbol{r}\|_2^2/(NL\sigma^2)$, which should be close to one for a good fit; values well above one suggest model error or a poor local minimum, and values well below one suggest over-estimated noise or over-fitting.

4.5 Relation to the bound

The bound is a local property of the statistical model and the scenario, while the estimator is an algorithm run on one noise realization. Comparing the estimator with the bound therefore requires many Monte Carlo runs under identical settings; the local bound at the estimate is useful for spotting weak directions but does not replace an RMSE-vs-CRLB comparison.

5 Simulation Results

5.1 Common settings

Two independent sets of outputs were produced: one for the hybrid bound and one for the hybrid MAP/ML estimator. They share the settings in Table 1. The observation window spans -77.5 to 77.5 ms and gives $NL = 4128$ complex samples. Pilots are unit-amplitude QPSK, the true path gains are $\boldsymbol{\beta} = [0.98877 + j0.14944,\ 0.45286 - j0.46628,\ 0.08498 + j0.49272,\ -0.23540 - j0.32340]^{\mathsf{T}}$, and the random seed is 42 in all runs.

Table 1: Main simulation settings.

Parameter	Value	Parameter	Value
K (RISs / reflected paths)	3	σ_p (RIS position std, per axis)	0.1 m
N (subcarriers, $-64 \dots 64$)	129	σ_v (RIS velocity std, per axis)	0.05 m/s
L (snapshots)	32	η (true)	2 ppm
Δf	480 kHz	δ_c (true)	40 ns
T_s	5 ms	Geometry model	reference-linearized
SNR (default)	20 dB		

The two scenarios differ. The bound uses $\boldsymbol{q} = [0, 0, 5]^{\mathsf{T}}$ m and $\boldsymbol{u}_0 = [8, 7, 1.5]^{\mathsf{T}}$ m, while the estimator uses $\boldsymbol{q} = [3, 4, 5]^{\mathsf{T}}$ m and $\boldsymbol{u}_0 = [8, 10, 1.5]^{\mathsf{T}}$ m. In both, the RISs are at $(0, 18, 3)$, $(20, 0, 4)$ and $(18, 18, 6)$ m with zero velocity. Because the CRLB depends strongly on geometry, the estimator error is *not* plotted on the bound curve of the other scenario; each set is analysed in its own geometry, and the estimator is compared only with the local bound saved in the same run.

5.2 Hybrid bound results

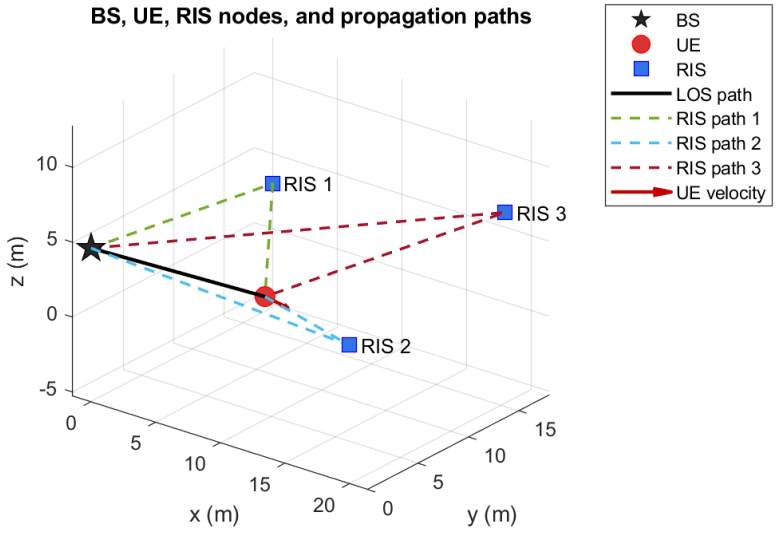


Figure 1: Geometry of the BS, the UE and the three RISs in the bound scenario. Solid line: direct path; dashed lines: RIS paths.

The three RISs lie in different directions from the UE and provide good diversity in the horizontal plane, while height differences are small and give little diversity in the vertical direction (Fig. 1). As shown next, this makes the z-axis parameters harder to estimate.

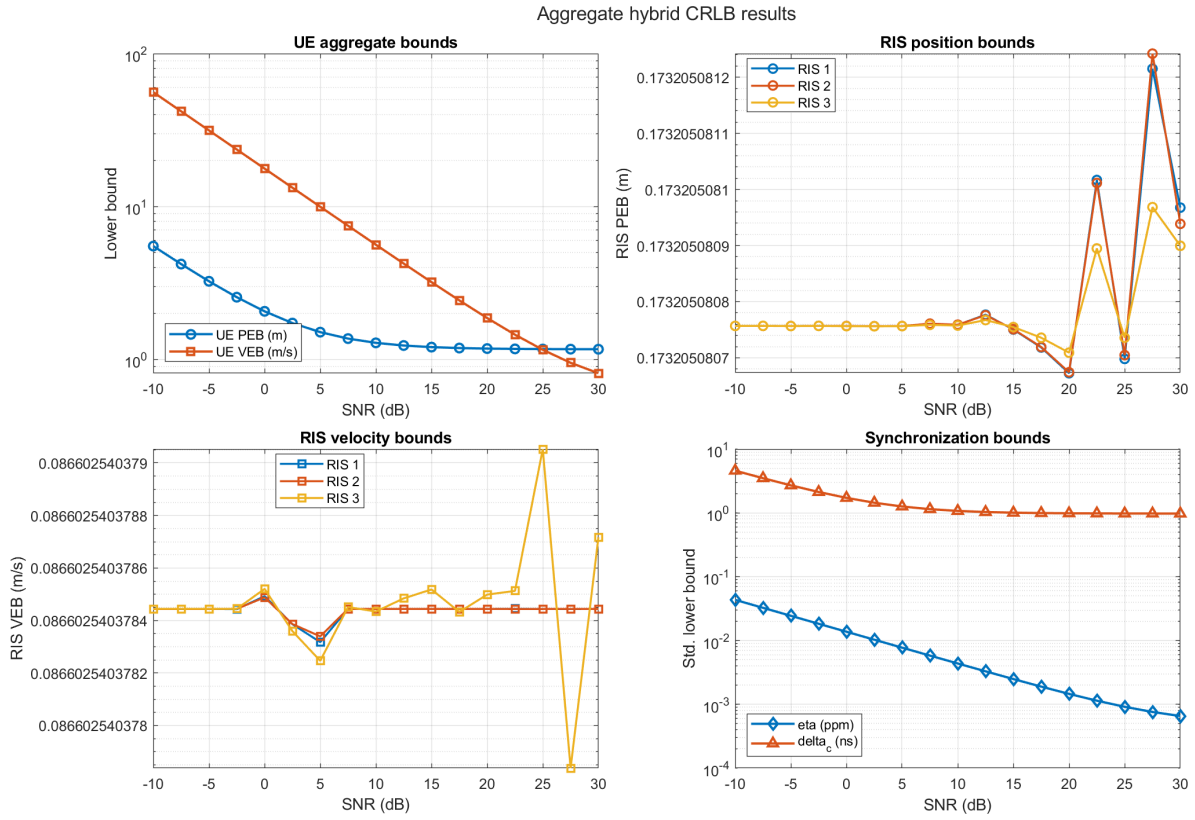


Figure 2: Aggregate bounds of the UE, the RISs and the synchronization parameters versus SNR.

Bounds versus SNR. Over -10 to 30 dB (Fig. 2), the UE PEB falls from 5.506 m to 1.165 m but flattens to an *error floor* of about 1.16 m at high SNR; the clock-offset bound similarly falls

from 4.586 ns to about 0.976 ns and saturates. In contrast, the UE VEB (55.94 to 0.811 m/s) and the η bound (0.04334 to 0.000655 ppm) keep decreasing. Time diversity from many snapshots is very effective for rate-type parameters, whereas for some position and timing parameters the RIS geometric uncertainty and the coupling between delays prevent unlimited improvement with more power. The tiny fluctuations of the RIS curves (order 10^{-9}–10^{-8}) are numerical, not physical.

Table 2: Hybrid bounds at SNR = 20 dB (`summary_bounds.csv`).

Quantity	Bound (std / vector error)	Unit
UE position (PEB)	1.17623	m
UE velocity (VEB)	1.86396	m/s
Position of each RIS	0.173205	m
Velocity of each RIS	0.086603	m/s
Relative frequency error η	0.001456	ppm
Clock offset δ_c	0.985476	ns

The RIS position bound equals $\sqrt{3}\sigma_p = 0.173205$ m and the RIS velocity bound equals $\sqrt{3}\sigma_v = 0.086603$ m/s (Table 2): in this configuration the observations are not enough to noticeably reduce the RIS uncertainty, and the prior dominates these blocks.

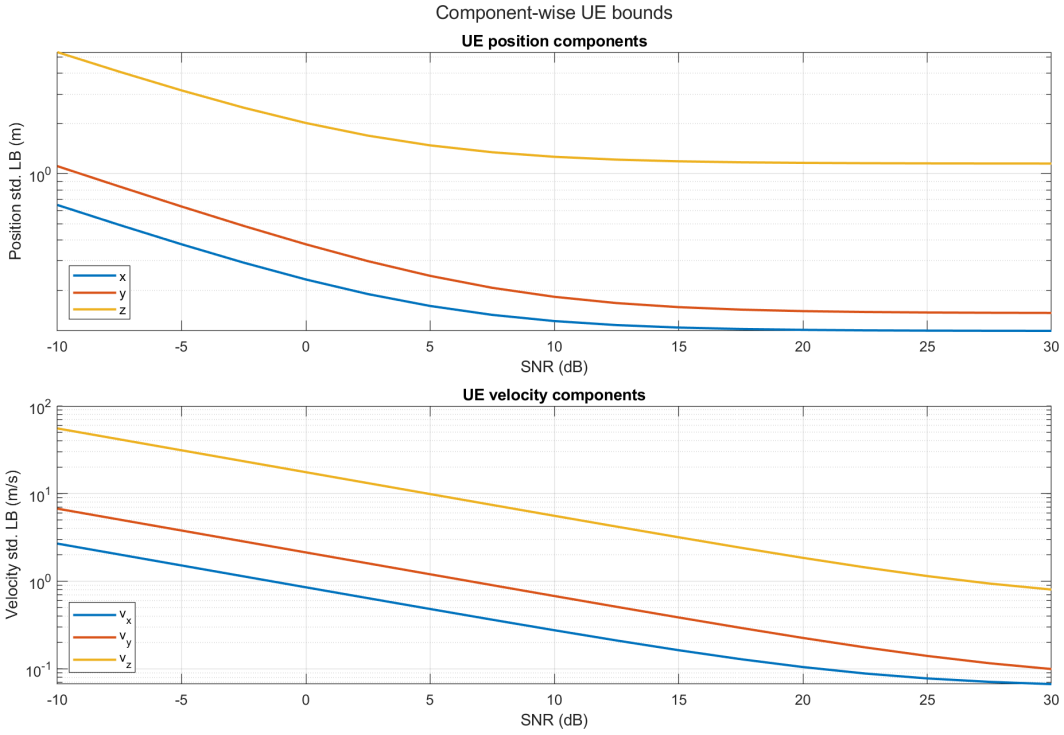


Figure 3: Per-axis bounds of UE position and velocity.

Per-axis UE bounds. At 20 dB the position bounds along x, y, z are about 0.116, 0.150 and 1.161 m, and the velocity bounds are about 0.105, 0.225 and 1.847 m/s (Fig. 3). The PEB is therefore dominated by the vertical axis — a geometric property of the scenario that higher SNR alone cannot fix. Per-axis RIS bounds stay at $\sigma_p = 0.1$ m and $\sigma_v = 0.05$ m/s and are almost flat in SNR. For the channel gains, the imaginary-part bounds of the first two reflected paths (about 0.01195 and 0.01045 at 20 dB) are larger than others: the same number of observations does not mean the same information about each path.

Rank and conditioning. At the default point the equivalent physical FIM has rank 26/26 and the full FIM rank 34/34, so no pseudoinverse was needed for rank deficiency. The condition number, however, is about 9.25×10^7. The eigenvalue spread and the correlation structure (Fig. 4) show strong coupling among some geometric, timing and channel parameters, which is one of the main reasons the vertical UE component is poorly estimated.

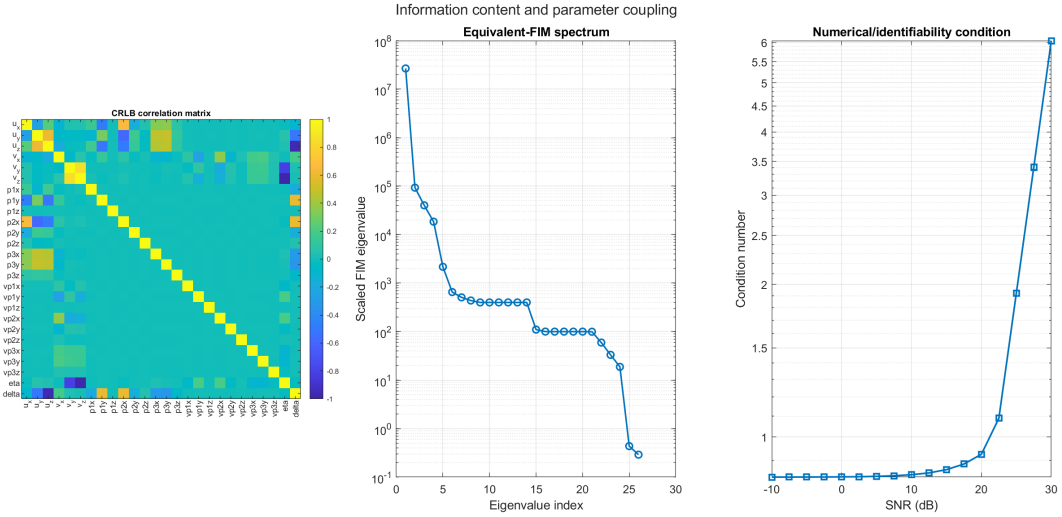


Figure 4: Bound correlation matrix, FIM eigenvalue spectrum and identifiability indicators.

Sensitivity to the RIS prior. The RIS position prior std was swept from 0.02 to 0.5 m, with the velocity std scaled proportionally from 0.01 to 0.25 m/s (Table 3, Fig. 5). The UE PEB grows almost linearly with σ_p, from 0.288 m to 5.822 m, and the VEB from 1.773 to 3.431 m/s. The RIS uncertainty therefore enters the localization error budget directly: the prior is not just a regularization term, its quality decides the final accuracy.

Table 3: Sensitivity of the UE bounds to the RIS prior (`prior_sensitivity.csv`).

σ_p (m)	σ_v (m/s)	UE PEB (m)	UE VEB (m/s)
0.02	0.010	0.288	1.773
0.05	0.025	0.606	1.793
0.10	0.050	1.176	1.864
0.20	0.100	2.334	2.124
0.50	0.250	5.822	3.431

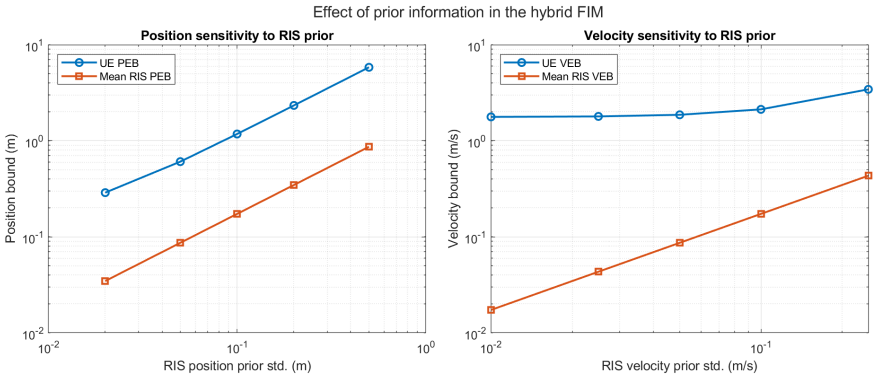


Figure 5: Sensitivity of the UE position and velocity bounds to the RIS prior standard deviation.

5.3 Hybrid MAP/ML estimator results

Geometry and user parameters. The prior mean of the RIS positions has small offsets of a few centimetres from the true positions, so it is fairly accurate. In Fig. 6, the final UE estimate moves from the initial point towards the true position, while the estimated RIS positions stay close to the prior mean — consistent with the bound result that the prior dominates the RIS states. The data are used mainly to improve the UE estimate rather than to move the RISs.

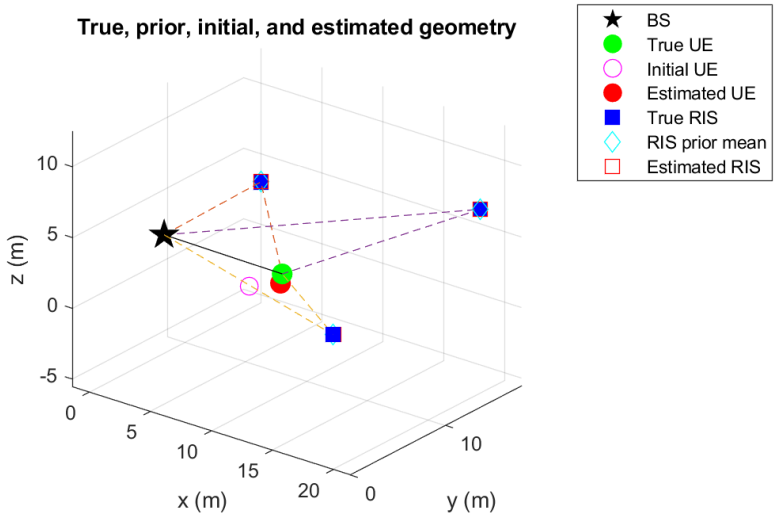


Figure 6: True geometry, prior mean, initial point and final estimate.

Table 4: UE parameters in the estimator run (`physical_parameters.csv`).

Parameter	True	Initial	Estimate	Error	Local std bound
u_x (m)	8.000	7.800	8.0185	0.0185	0.1373
u_y (m)	10.000	6.800	9.8000	−0.2000	0.2194
u_z (m)	1.500	1.700	0.9162	−0.5838	1.0483
$\dot{u}_x$ (m/s)	1.200	1.000	1.3672	0.1672	0.3112
$\dot{u}_y$ (m/s)	−0.400	−0.300	−0.4696	−0.0696	0.1906
$\dot{u}_z$ (m/s)	0.100	0.000	0.3569	0.2569	0.8496

The vector UE position error drops from 3.212 m at the initial point to 0.617 m, an improvement of about 80.8% (Table 4). The x component is the most accurate (0.0185 m error); the height error (−0.5838 m) contributes most of the final error, consistent with its large local bound (1.0483 m) and with the weak vertical direction found in the bound analysis.

Active local bound. The estimate of u_y is exactly 9.8 m, i.e. on the upper edge of the local box $(6.8 + 3$ m), while the true value 10 m lies outside it. Part of this error is therefore caused by the choice of the local search region, not only by noise or limited information.

The velocity error grows from about 0.245 m/s at the initial point to 0.314 m/s. This does not mean the estimator fails: the bound describes average performance over many noise realizations and does not require every subset of parameters to improve in a single run. It does, however, show that the velocity estimate needs a Monte Carlo evaluation.

Synchronization initialization. The grid search reduces the initial cost from 407 890 to 106 313, moving the starting point from $(\eta, \delta_c) = (1$ ppm, 30 ns) to (2 ppm, 35 ns) (Fig. 7). The cost surface is elongated along δ_c, so the clock offset is less separable at this stage, while the cost changes

faster along η thanks to the time information from many snapshots. Local optimization then gives $\hat{\eta} = 1.999864$ ppm and $\hat{\delta}_c = 40.4181$ ns, i.e. absolute errors of 0.000136 ppm and 0.418 ns.

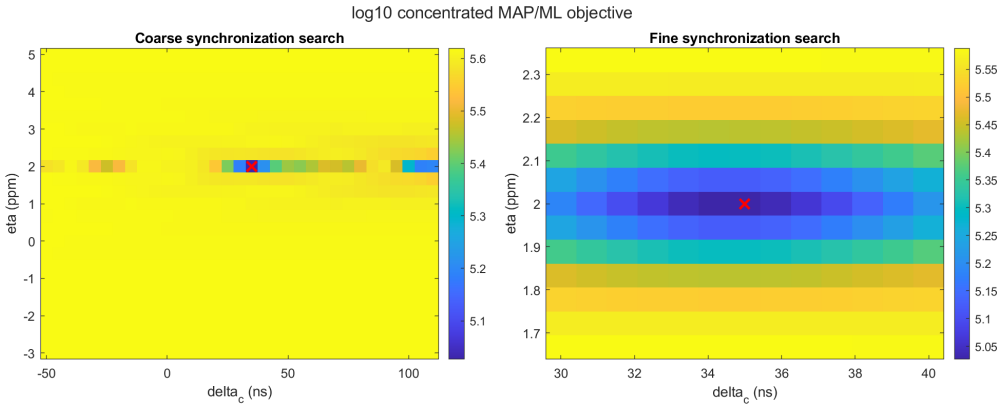


Figure 7: Concentrated cost surface in the coarse and fine synchronization search.

Convergence, residual and channel gains. All eight selected starts converge to almost the same final cost (4123.258); the best start stopped after about 80 iterations with exit flag 2 (step size below tolerance, constraints satisfied). The residual map shows no noticeable structure after the fit (Fig. 8). The data part of the cost is 4123.136 and the prior part only 0.1218, so the normalized residual is $4123.136/4128 = 0.99882$, consistent with the assumed noise variance. Convergence of all starts to the same value indicates a stable solution inside the chosen search region, though it does not prove global optimality.

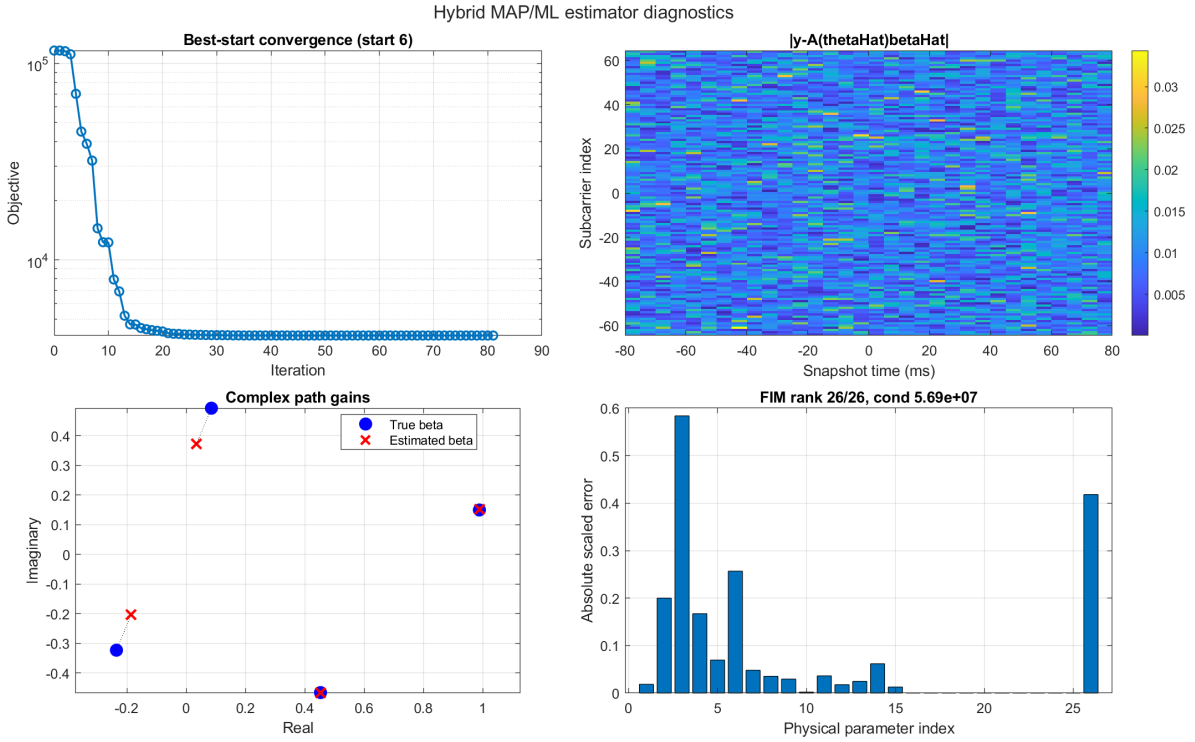


Figure 8: Estimator diagnostics: convergence of the best start, residual map, complex path gains and absolute scaled error of the physical parameters.

Table 5: True and estimated complex path gains (`channel_parameters.csv`).

k	True β_k	Estimated $\hat{\beta}_k$	Absolute error
0	$0.98877 + j0.14944$	$0.98984 + j0.15166$	0.00247
1	$0.45286 - j0.46628$	$0.45206 - j0.46788$	0.00179
2	$0.08498 + j0.49272$	$0.03470 + j0.37246$	0.13036
3	$-0.23540 - j0.32340$	$-0.18601 - j0.20389$	0.12932

The gains of the first two paths are estimated very accurately, while paths 2 and 3 have absolute errors of about 0.13 (Table 5). This agrees with the information analysis (some gain components carry less information) and with the coupling between path gains, geometry and synchronization, through which part of the geometric error can leak into the gains. At the estimate, the local equivalent FIM has rank 26/26 and condition number 5.69×10^7: locally identifiable, but with very uneven information across directions.

5.4 Summary of the numerical analysis

1. The quality of the RIS prior directly limits UE localization: increasing the RIS position uncertainty can multiply the UE PEB, so the accuracy of the RIS geometry is part of the fundamental localization limit.

2. With three RISs and 32 snapshots the physical FIM is full rank, but its condition number is of order 10^7–10^8; the vertical component of UE position and velocity is one of the weakest directions.

3. Analytic removal of the gains, grid initialization of the synchronization parameters and multi-start optimization improve the solution markedly over the initial point (3.212 m $\rightarrow$ 0.617 m), and all eight starts reach practically the same cost.

4. The local bound on u_y was active in the final solution, so the local search region — especially along y — should be enlarged, or the problem re-solved with a larger region whenever a bound becomes active.

5. The estimator results come from a single noise realization; a proper comparison with the bound requires identical scenarios and Monte Carlo RMSE at several SNR values.

6 Conclusions and Future Work

6.1 Conclusions

This project studied the localization and synchronization of a mobile UE aided by RISs whose geometry is uncertain. Instead of treating this uncertainty as an external disturbance, it was built into the statistical model, the bound and the estimator: UE, synchronization and channel parameters were modelled as deterministic unknowns, and the RIS states as random parameters with a prior. The OFDM model showed that geometric delay, motion, frequency error and clock offset act jointly on the phase, so subcarrier (frequency) diversity and snapshot (time) diversity are complementary sources of information. The hybrid FIM combined data and prior information; the channel gains were removed with a Schur complement while the uncertain RIS states were kept, preserving the effect of their uncertainty in the final bound. At 20 dB, the bound gave a UE PEB of 1.176 m and VEB of 1.864 m/s, with the vertical axis weakest. The prior sensitivity study showed that the quality of the RIS map is part of the error budget: in practice, better calibration and RIS localization can matter as much as more power or more observations. The hybrid MAP/ML estimator, with variable projection, coarse-to-fine synchronization search, multi-start and constrained local optimization, reduced the UE position error from 3.212 m to 0.617 m

with a normalized residual close to one. These results describe the behaviour, sensitivities and practicality of the method rather than a final claim that the estimator attains the bound.

6.2 Limitations

1. **Constant-velocity motion** — not valid under acceleration or sudden manoeuvres.
2. **ICI neglected** — may miss part of the data structure for large frequency errors, strong Doppler or fast-varying channels.
3. **Constant channel gains over the block** — real gains may change with motion, blockage or RIS reconfiguration.
4. **Different geometries in the two output sets** — direct comparison of estimator error with the other scenario's bound is not valid.
5. **Single noise realization** — not enough for statistical conclusions; RMSE over many realizations is needed.
6. **Local optimization and search bounds** — no guarantee of a global minimum, and the active local bound on u_y is partly responsible for its error.
7. **Limited geometries** — rank, condition number and bounds can change considerably with BS, UE and RIS placement and with the number of paths.

6.3 Future work

1. Fully align the bound and estimator scenarios and run Monte Carlo trials at several SNRs to compare RMSE directly with the CRLB.
2. Extend the motion model with acceleration (re-deriving geometric derivatives and re-checking identifiability).
3. Model ICI explicitly and determine the range of speeds and frequency errors in which the current approximation holds.
4. Use a time-varying channel model (piecewise-constant, stochastic or low-dimensional in time) for $\boldsymbol{\beta}$.
5. Study more RISs and more diverse geometries, and design RIS placement using PEB, VEB, the smallest eigenvalue or the condition number.
6. Extend to multiple users (shared RISs, interference, pilot allocation, and joint use of multi-user data for RIS calibration).
7. Adapt the local search region automatically: re-solve with a larger region when a local bound becomes active, and compare with more global strategies or geometry-based initialization.
8. Study robustness to model errors in σ^2, σ_p, σ_v and to unmodelled paths.

A Parameter Ordering and Output Files

Table 6: Indices of the physical parameters in $\boldsymbol{\theta}$ for $K = 3$.

Index	Parameter	Unit
1–3	UE initial position $\boldsymbol{u}_0$	m
4–6	UE velocity $\dot{\boldsymbol{u}}$	m/s
7–9, 10–12, 13–15	RIS 1, 2, 3 positions $\boldsymbol{p}_{k,0}$	m
16–18, 19–21, 22–24	RIS 1, 2, 3 velocities $\dot{\boldsymbol{p}}_k$	m/s
25	relative frequency error η	–
26	clock offset δ_c	s
27–30	$\mathrm{Re}\{\beta_0\}, \ldots, \mathrm{Re}\{\beta_3\}$	–
31–34	$\mathrm{Im}\{\beta_0\}, \ldots, \mathrm{Im}\{\beta_3\}$	–

Table 7: CSV outputs of the two packages.

File	Content
Bound package (`crlb/crlb_results/`)	
`summary_bounds.csv`	aggregate UE, RIS and synchronization bounds at the default SNR
`all_parameter_bounds.csv`	variance and std lower bound of each of the 34 parameters
`snr_sweep_bounds.csv`	bounds over -10 to 30 dB
`prior_sensitivity.csv`	PEB and VEB versus σ_p and σ_v
Estimator package (`estimator/hybrid_map_results/`)	
`summary.csv`	aggregate errors, cost components, rank and condition number
`physical_parameters.csv`	true, initial, estimated value, error and local bound per parameter
`channel_parameters.csv`	true and estimated complex path gains
`multi_start_summary.csv`	initial and final cost and exit flag of each start
`sync_initialization.csv`	result of the grid search over η and δ_c

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